# Chapter 7 Forming and Shaping of Glass

**Molten Glass**

**Flat Sheet and Plate Glass**

Float Method- Molten glass is floated over a “bath” of molten tin before it is solidified in a separate chamber.

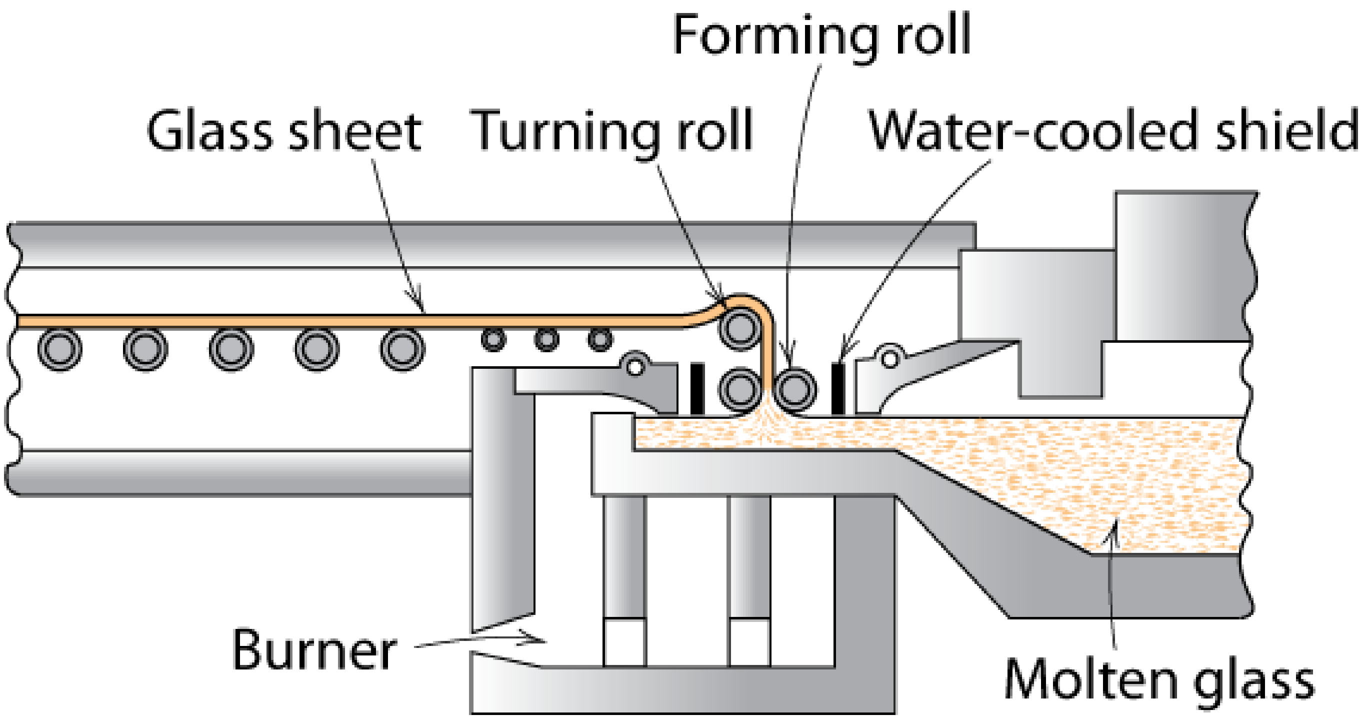
No additional finishing is necessary.

Drawing Process- Molten glass is squeezed through two rolls, then moves on o two smaller rolls.

Rolling Process- Similar to drawing process, but patterns are commonly imprinted from the rolls onto the glass, leaving a rough finish.

# Sheet Glass Forming

- Sheet forming – continuous draw
  - originally sheet glass was made by "floating" glass on a pool of mercury – or tin




Adapted from Fig. 13.9, *Callister 7e.*

# Glass Tubing and Rods

**Tubing-** Molten glass is wrapped around a mandrel and taken out by two rolls.

Air is blown through the mandrel to prevent the tube from collapsing into itself.

Some machines manufacture 2000 fluorescent light bulbs per minute using this method.

**Rods-** Rods are made in exactly the same way,

but without the air blown through

the mandrel.

This allows the glass to collapse and

become solid.

# Discrete glass products

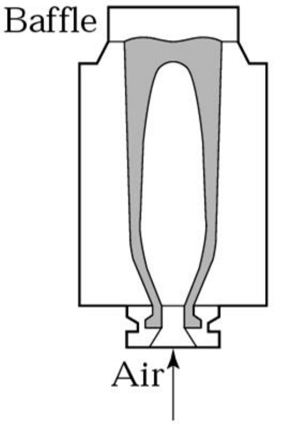
Processes used to make discrete glass objects

- Blowing
- Pressing
- Centrifugal casting
- Sagging

# Blowing

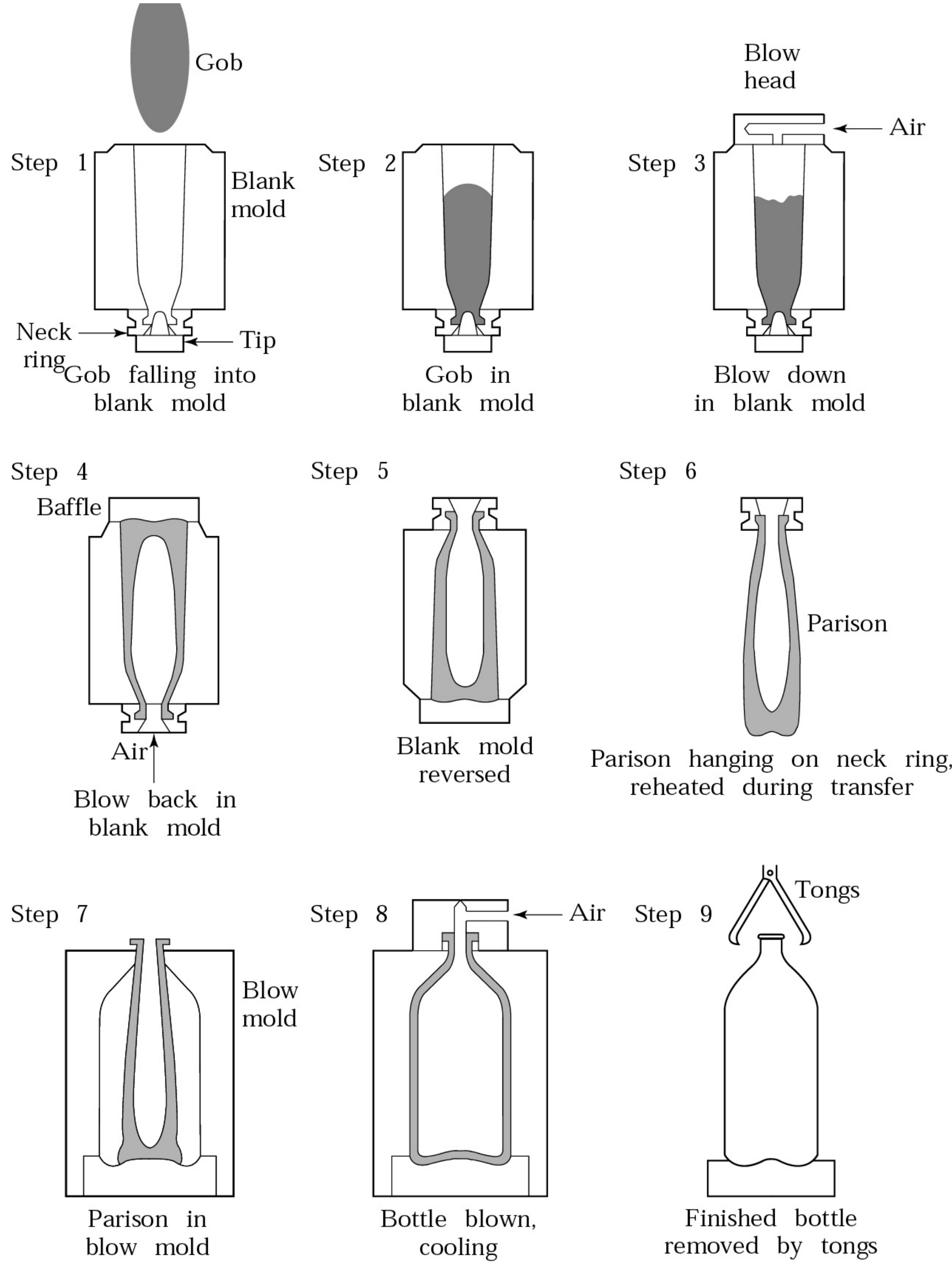
- Blowing process: Blown air expands a hollow gob of heated glass against the inner walls of a mold.

- A parting agent (such as oil or emulsion) is usually used to prevent the glass from sticking to the mold.



# Blowing

- Blow and blow process: After blowing a second blowing operation can be used for finalizing product shape.



# Pros and cons of blowing

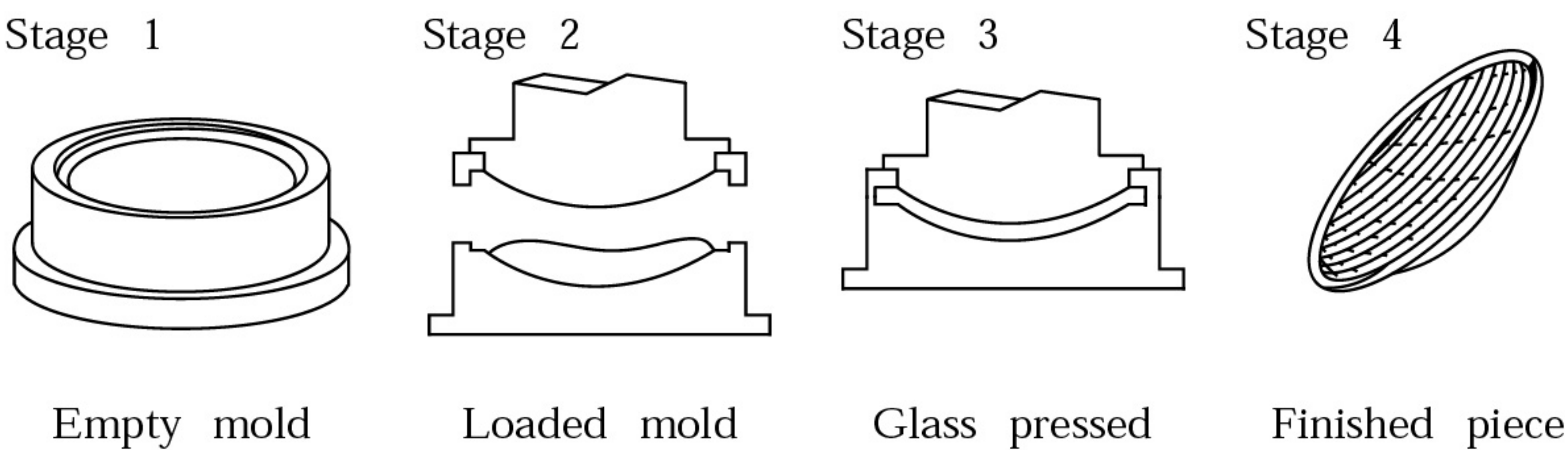
- Pros: Very economical for high-rate production.

  Example: Highly-automated blowing machines can make around 2000 incandescent light bulbs per minute.

- Cons: Difficult to control the wall thickness of the product

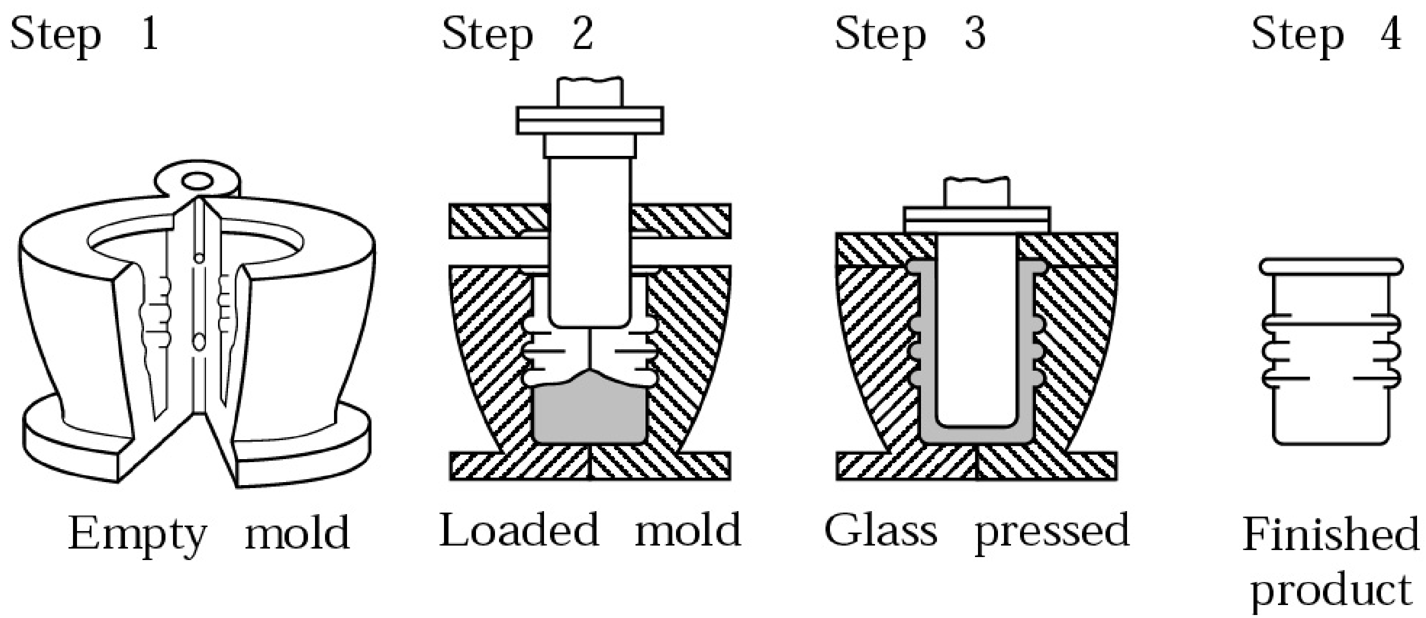
# Pressing

- Pressing process: A gob of molten glass is placed into a mold and pressed by a plunger into a confined shape.
- Molds may be one piece or split. Solidifying glass acquires the shape of the mold-plunger cavity.
- Similar to closed-die forging.



# Pressing

- One-piece molds cannot be used in pressing if the plunger cannot be retracted.
- One-piece molds cannot be used for thin-walled items
- Split molds can accommodate thin-walled products



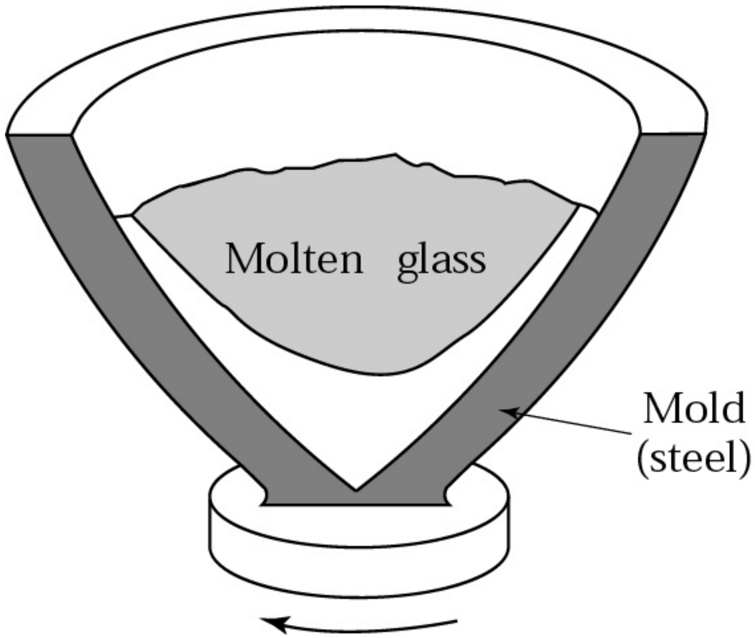
# Pressing

- Pressing can produce higher dimensional accuracy than blowing.
- After a part is pressed, it is blown to further expand the glass into the mold.

# Centrifugal Casting (Spinning)

- Centrifugal casting process: The centrifugal force pushes the molten glass against the wall.

- TV picture tubes and missile nose cones can be made with centrifugal casting.



# Sagging

- Sagging process: A sheet of glass is placed over a mold and heated. The glass sags by its own weight and takes the shape of the mold.

- Typical applications include dishes, sunglass lenses, mirrors for telescopes, and lighting panels.

# Glass ceramics manufacture

- Trade names: Pyroceram, Corning ware
- Contain large proportions of several oxides.
- Manufacturing involves a combination of methods used for ceramics and glasses.
- Shaped into discrete products (such as dishes and baking pans) then heat treated.
- After heat treating glass is devitrified (recrystallized).

# Glass Fibers

- Continuous glass fibers are drawn through multiple orifices (200 to 400 holes) in heated platinum plates at speeds as high as 500 m/s.
- Fiber diameters as small as 2 $\mu$m.
- Coated with chemicals to protect fiber surface.
- Short fibers (chopped) are made as compressed air or steam passes the fiber as it passes through the orifice.

# Glass Fibers - Glass wool

- Glass wool is short glass fibers.

- Glass wool is used for thermal and acoustic insulation.

- Made by a centrifugal spraying process. Molten glass is ejected (spun) from a rotating head.

- Glass wool fiber diameter is typically 20 to 30 μm.

# Techniques for Strengthening and Annealing Glass

- Glass can be strengthened by thermal tempering, chemical tempering, and laminate strengthening.

- Finishing operations can be used to impart desired properties and surface characteristics.

# Thermal Tempering

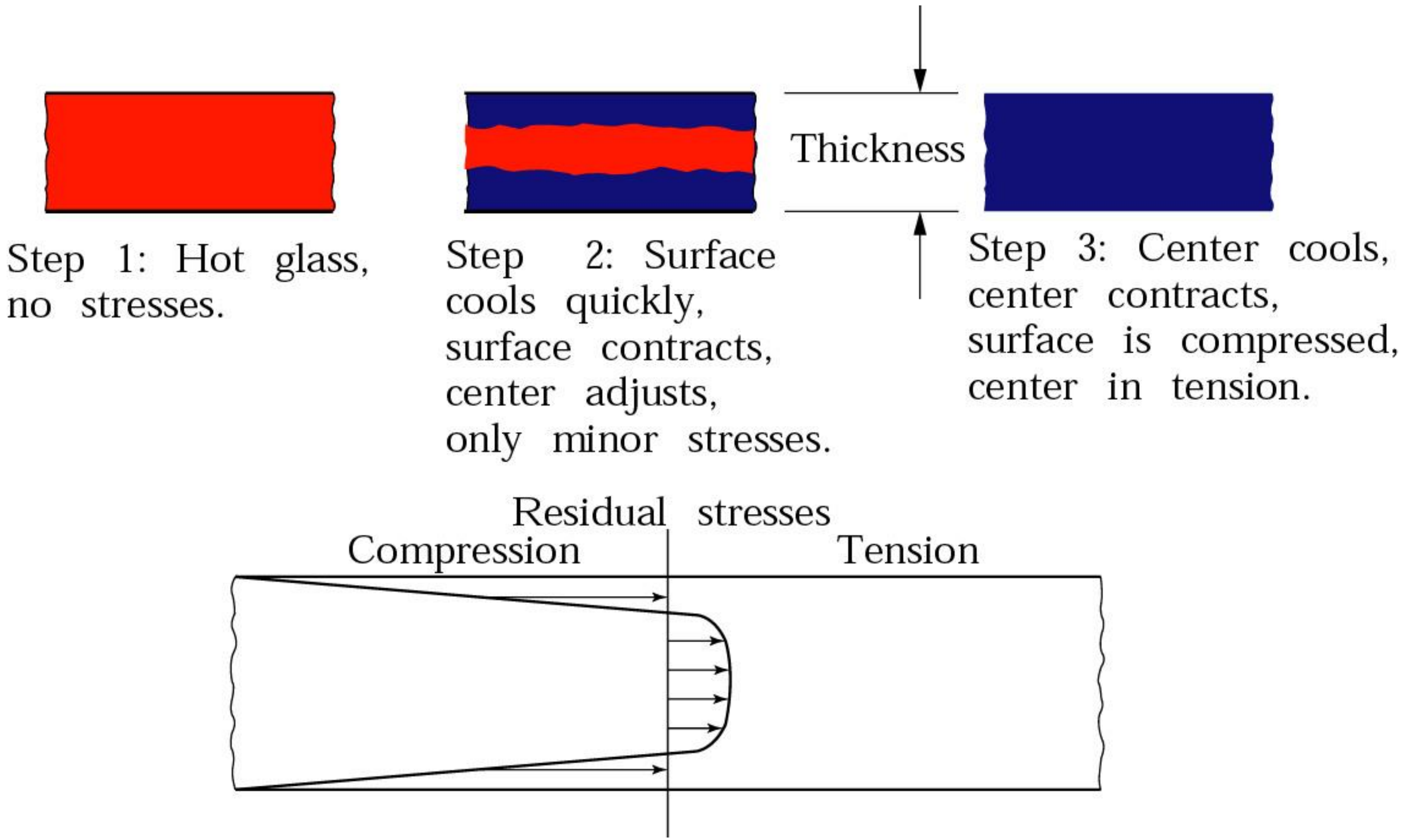
- Surfaces of the hot glass are cooled rapidly by a blast of air. The surfaces solidify and are forced to contract as the bulk of the glass begins to cool.

- Surfaces develop residual compressive stresses.

- The interior develops tensile stresses.

- Compressive surface stresses improve the strength of the glass.

# Thermal Tempering



# Chemical Tempering

- The glass is heated in a bath of molten $KNO_3$, $K_2SO_4$, or $NaNO_3$, depending on the type of glass.
- Ion exchanges take place and larger atoms replace smaller atoms on the surface of the glass.
- Residual compressive stresses develop on the surface.

# Laminated Glass

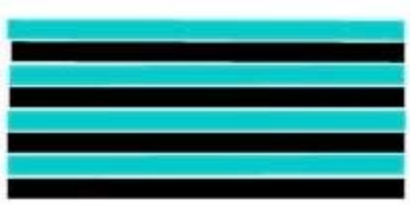
- Glass is strengthened through a method called laminate strengthening.

- Two pieces of flat glass have a thin sheet of tough plastic in between.

- When the glass is cracked, its pieces are held together by the plastic sheet.

# Bulletproof Glass

- Bulletproof glass basically consists of glass laminated with a polymer sheet.
- Thickness ranges from 7 to 75 mm. Thinner glass is for handguns and the thicker glass is for rifles.



# Finishing Operations

- Annealing removes residual stresses by heating the glass to a certain temperature and then cooling gradually.
- Annealing time ranges from a few seconds to 10 months.
- Glass products may be cut, drilled, ground, and polished.
- Care should be exercised in all finishing operations to ensure there is no surface damage.

# Design Considerations for Ceramics and Glasses

- Ceramic and glass products require careful selection of composition, processing methods, finishing operations, and methods of assembly with other components.